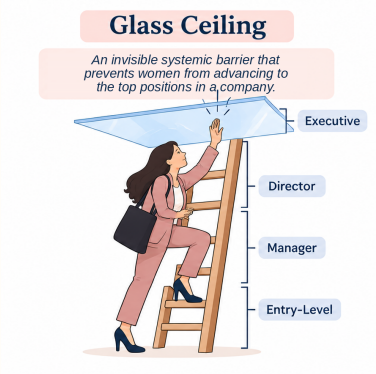
Board Diversity and the Career Progression of Women: Evidence from SB826

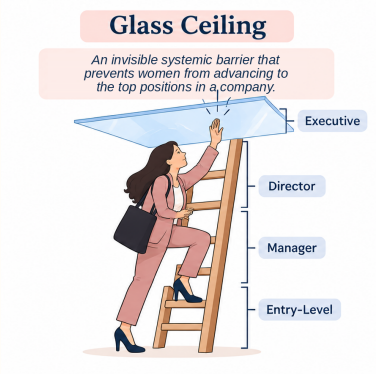
Anushri Jain
Washington University in St.Louis

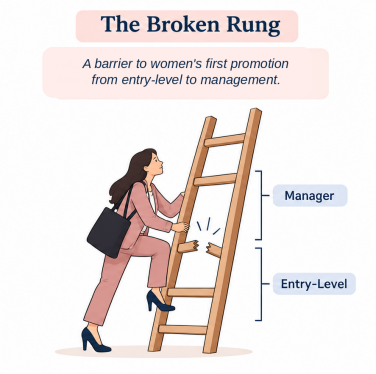
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Barriers to Women's Career Progression

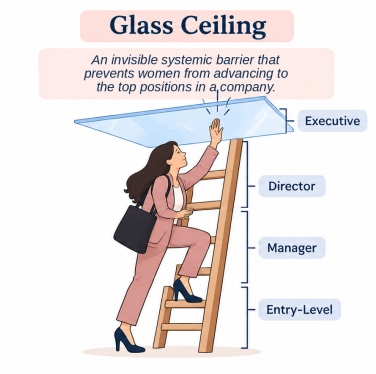


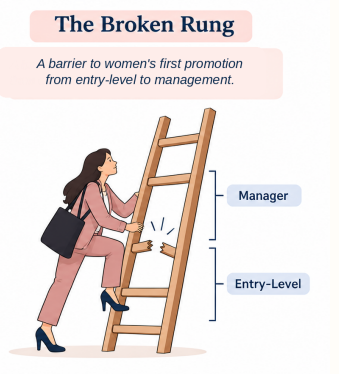
Barriers to Women's Career Progression





Barriers to Women's Career Progression





In a recent survey (*McKinsey & Company*), women identified **advancing from entry-level to mid-management** as their biggest career challenge.

Our Research Questions

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- How do these changes vary across organizational levels?
- Through what channels does greater female representation on boards influence women's career opportunities within the firm?

Why Women in Leadership May Facilitate Career Progression



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Why Women in Leadership May Facilitate Career Progression









And Why It Might Not

- **Tokenism:** Women may be appointed solely to meet quota requirements, potentially limiting their influence.

And Why It Might Not

- **Tokenism:** Women may be appointed solely to meet quota requirements, potentially limiting their influence.
- **Backlash:** Firms may respond by hiring fewer women elsewhere in the organization.

Answering These Questions is Challenging

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- Board **appointments are endogenous**, often reflecting firms' existing attitudes toward gender diversity and female employees.
- Most board gender quotas are **implemented at the national level**, making it difficult to identify suitable treatment and control groups.
- Board quotas often have **lengthy implementation periods**, making it difficult to pinpoint the timing of the policy shock.

Identification Strategy - Passage of SB 826

- **Treatment Group:** California firms with all-male boards in 2017.

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- **Well-defined policy timeline:** SB 826 was introduced and passed in 2018, with initial compliance required by the end of 2019. Timeline
- **Difference-in-differences:** Compare changes in women's career outcomes between treated and control firms before and after the quota.

Difference-in-Differences Specification

$$Y_{i,t} = \beta_0 + \beta_1, \text{Post}_t \times \text{CA}_i + \alpha_i + \gamma_t + \varepsilon_{it} \quad (1)$$

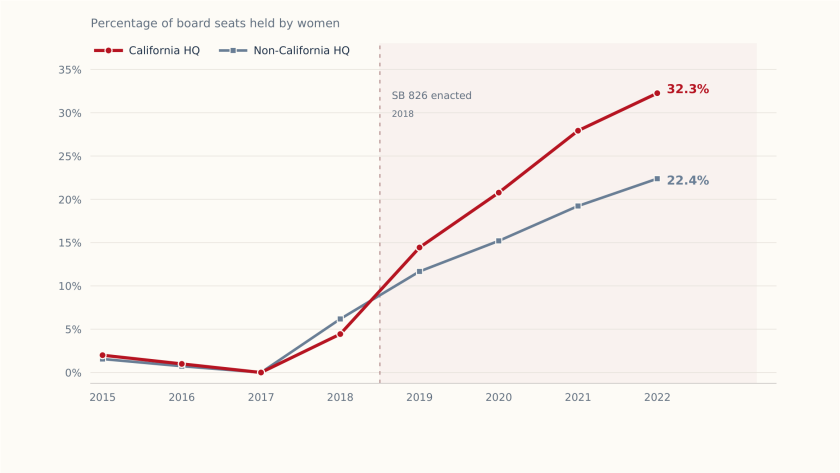
- **Key Variables:**
  - *CA*: Indicator for firms headquartered in **California** in 2017
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- **Parallel Trends Assumption:** In the absence of SB 826, women's career outcomes in treated and control firms would have followed similar trends over time.

Female Board Representation Following SB 826



Following SB 826, female board representation increased more rapidly among California-HQ firms.

Data and Career Outcomes

Data: LinkedIn employer–employee matched employment histories.

Gender: Gender API.

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- **Top Management:** C-suite and VPs *Keywords:* Chief", CFO", "Vice President"
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- **Entry Level:** All other positions *Examples:* Analyst", Associate"

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Workplace Mechanisms: Glassdoor ratings of workplace culture and career opportunities.

Graph

Results: Overall

Normalized by female headcount in 2017

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
CaliforniaHQ × Post2018	0.167**	0.103***	0.019**	0.017***	-0.000
	(0.065)	(0.025)	(0.008)	(0.005)	(0.003)
Firm FE	✓	✓	✓	✓	✓
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R^2	0.530	0.387	0.380	0.283	0.269
Observations	2,109	2,109	2,109	2,103	2,115

- **Economic Magnitudes (per firm): +8.9 women, +5.5 female hires, +1 promotion,** and **+0.9 departures**.

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Entry-Level Graph Mid-Management Graph

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CaliforniaHQ × Post2018	0.053	0.057	0.004	0.011*	-0.005**
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- **Economic Magnitudes (per firm): Entry Level:** +3 female hires, +0.6 female departures, and a 0.5 pp decline in exit rate. **Mid-Management:** +3 women employed, +1.8 female hires, and +0.5 female promotions.

Results: Top Management

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
CaliforniaHQ X Post2018	0.056	0.012	−0.000	0.003	0.004*
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- **Economic Magnitudes (per firm): +3 women in top management**, **+0.6 female hires**, **+0.2 female departures**, and **+0.4 percentage points in exit rate**.

Results: Top Management

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
CaliforniaHQ X Post2018	0.056	0.012	−0.000	0.003	0.004*
	(0.005)	(0.002)	(0.001)	(0.001)	(0.001)
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
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Mechanism: Supportive Workplace Policies

	Overall	Culture & Values	Comp. & Benefits	Career Opp.	Work–Life	Senior Leaders
CaliforniaHQ × Post2018	0.276	0.242	0.092	0.170*	0.182	0.279**
	(0.041)	(0.038)	(0.024)	(0.071)	(0.041)	(0.039)
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
R^2	0.410	0.421	0.440	0.361	0.395	0.395
Observations	1,398	1,362	1,362	966	1,363	1,361

Glassdoor employee ratings improve across all six dimensions (1–5 scale).

Stronger Effects In Firms with Female Directors on Committees

	Career Progression					Glassdoor
	Total	Hired	Promoted	Exit	Exit Rate	Overall Rating
CaliforniaHQ × Post2018 × Committee2019	0.404*	0.166**	0.085***	0.026**	0.011**	0.148*
	(0.217)	(0.074)	(0.030)	(0.013)	(0.005)	(0.081)
CaliforniaHQ × Post2018	-0.035	0.008	-0.032	0.001	-0.008*	0.175
	(0.202)	(0.072)	(0.024)	(0.014)	(0.003)	(0.083)
Post2018 × Committee2019	-0.213	-0.090	-0.005	-0.018	-0.010**	-0.018
	(0.217)	(0.074)	(0.030)	(0.013)	(0.005)	(0.080)
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
R^2	0.542	0.389	0.379	0.315	0.326	0.449
Observations	1,798	1,798	1,798	1,793	1,807	1,176

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- **There is only one treated state?** *Tests:* Bootstrap inference, jackknife estimation, and alternative clustering specifications.

Conclusion

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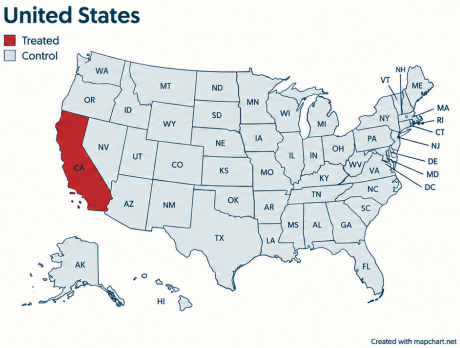
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  - **Across the organizational hierarchy:** Increases hiring and retention at the entry level, and hiring and promotions in mid-management.
  - **Mechanisms:** Female directors may improve workplace policies and internal career opportunities, consistent with improvements in employee ratings and stronger effects when women serve on board committees.

Supporting Slides

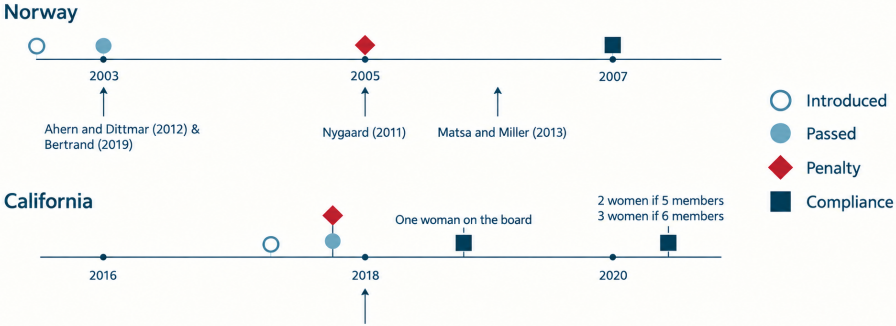
Identification





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Timeline

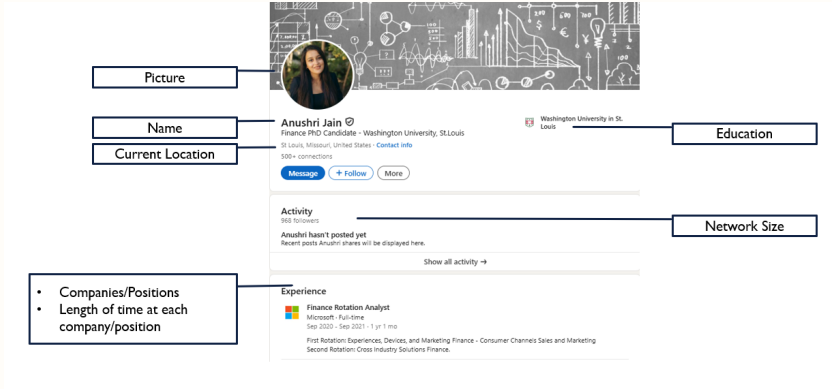


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Data Sources

- **BoardEx Data**
  - Use reported gender ratio to identify firms with **0 female board members.**
  - Merge with Compustat and Notre Dame SRAF to identify headquarters of firms.
  - Final sample: **183 treated firms** and **1,020 control firms.**
- **ExecuComp Data**
  - Merge BoardEx data with ExecuComp to identify *"top earners."*
  - Covers only S&P 1500 firms — **54 treated** and **210 control firms.**
- **LinkedIn Data**
  - Over **86,000 profiles.** Example

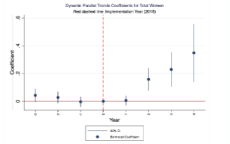
LinkedIn



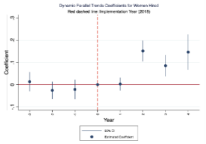
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Dynamic Plots: Overall

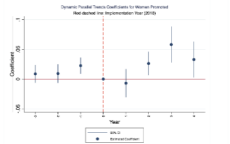
Figure 6: Dynamic Parallel Trends for Women's Career Outcomes



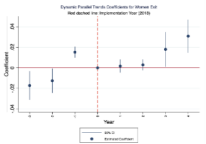
(a) Total Women Normalized by 2017 Women



(b) Hired Women Normalized by 2017 Women



(c) Promoted Women Normalized by 2017 Women Total

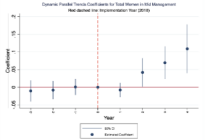


(d) Exiting Women Normalized by 2017 Women Total

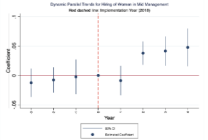
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Dynamic Plots: Mid-Management

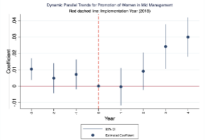
Figure 8: Dynamic Parallel Trends for Women in Mid Management Positions



(a) Total Women in Mid Management Normalized by 2017 Women



(b) Hired Women in Mid Management Normalized by 2017 Women

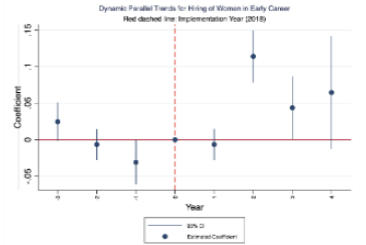


(c) Promoted Women Normalized by 2017 Women

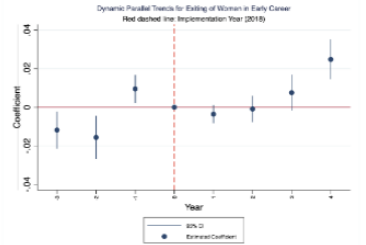
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Dynamic Plots: Early-Career

Figure 9: Dynamic Parallel Trends for Women in Early Career Positions



(a) Hired Women in Early Career Normalized by 2017 Women



(b) Exiting Women in Early Career Normalized by 2017 Women

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